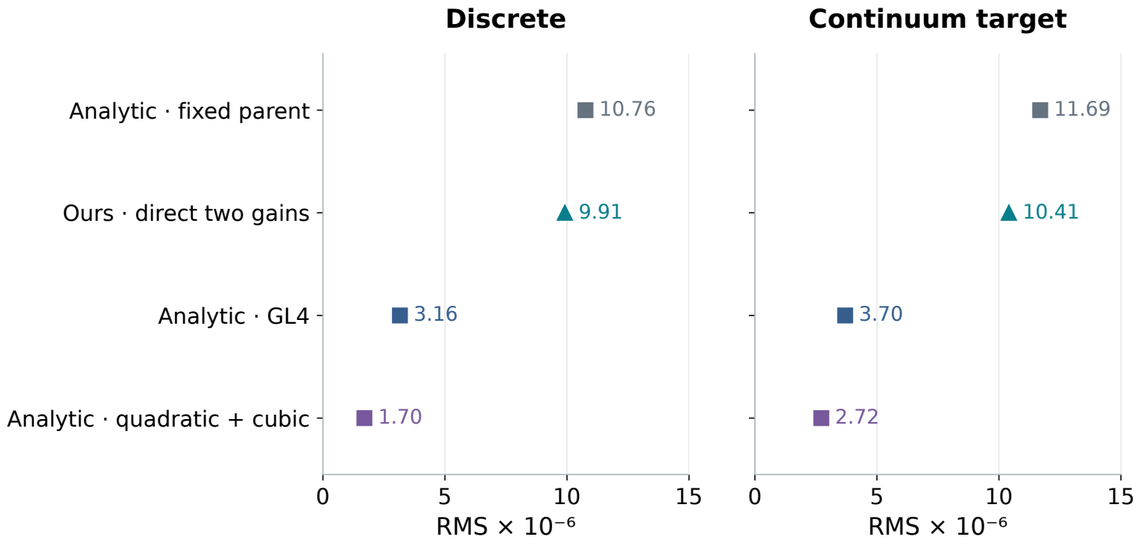
TDN Resolve: what the native run established

RTX 4090 native run • Four different research questions • Development evidence only

A Calibration versus a stronger formula

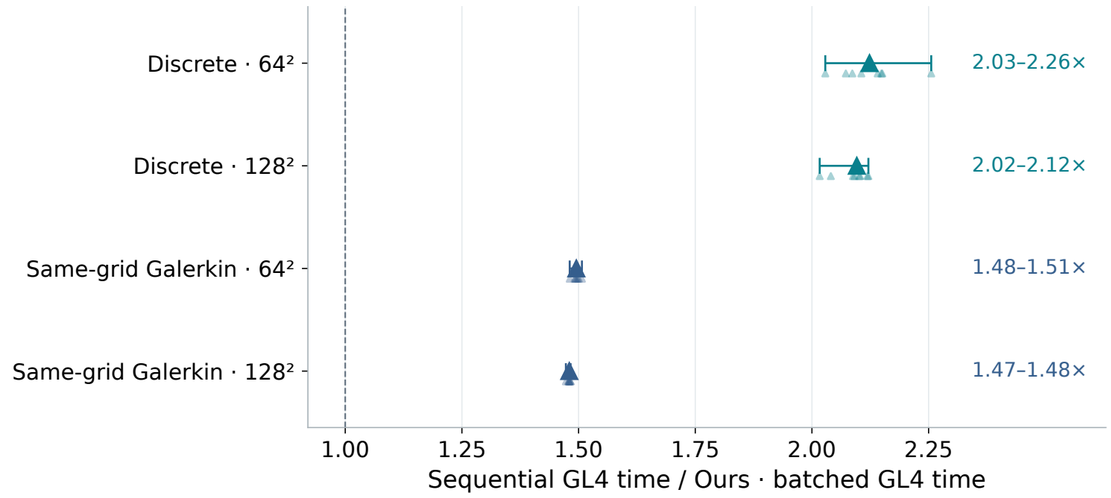
Lower RMS is better · same fit cohort; controls kept separate



32 validation fields × 4 horizons per track; equal-field mean RMS.
Small fitted gain versus the parent; GL4 and nested analytic terms remain stronger.

B GL4 execution: batching the same rule

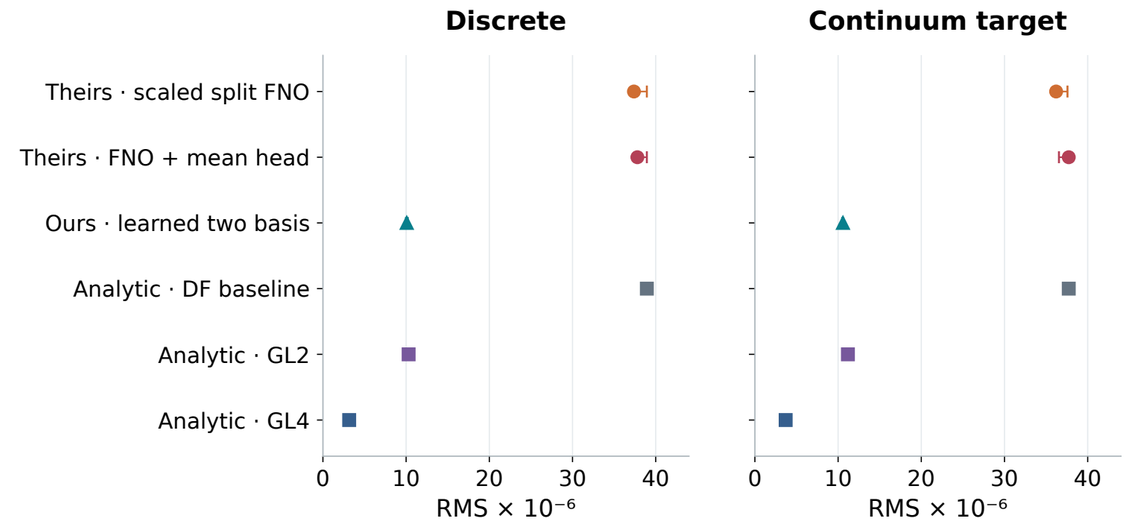
Higher speedup is better · complete warm calls, same output formula



8 conditions per row; 20 paired rounds per condition. Whiskers: observed min–max, not CI.
30/32 conditions meet both accuracy targets; maximum parity difference ≤ 1.79 × 10⁻⁷.

C Adapted FNO fairness checks

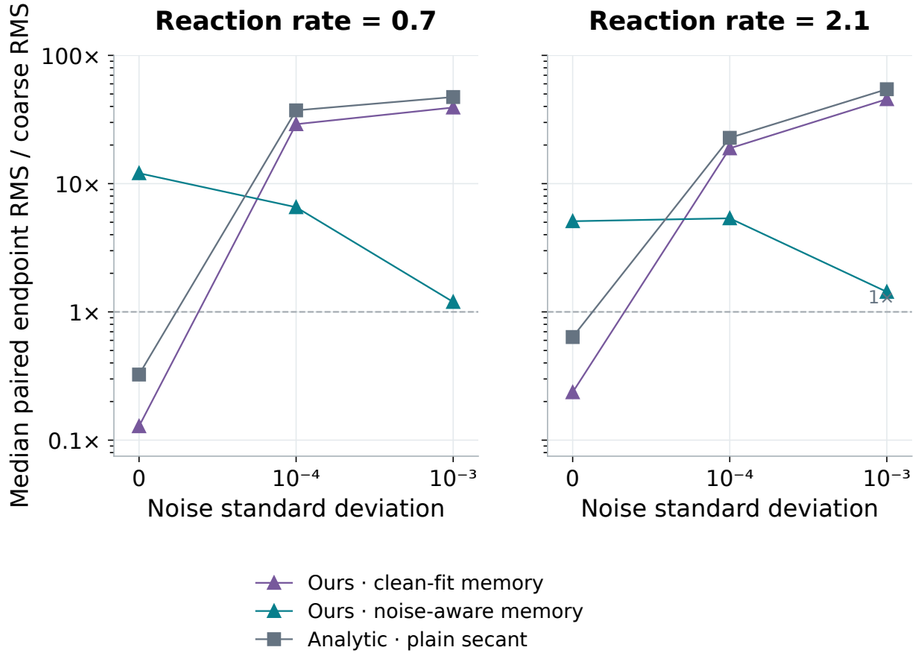
Lower RMS is better · validation-selected checkpoints, equal data



32 validation fields × 4 horizons; 3 training seeds. Whiskers: seed range, not CI.
Analytic controls repeat identical outputs. These are adapted implementations,
not a reproduced published-FNO benchmark or untouched confirmation.

D Memory: clean gains do not survive noise

Lower ratio is better · below 1 beats the coarse RK4 control



16 independent parents × 2 noise draws; 32-step autonomous rollout.
Coarse RK4 and validation-selected zero damping both equal 1×. No interval shown.

No universal leaderboard: panels use different tasks, cohorts and estimands. Raw values, denominators and SHA-256 source traces are in plotted-data.json.